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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

SE 19-206-0604 LIFE SAFETY IN BUILDING FIRE
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Plan and design evacuation route and exits based on the concepts of human behaviour under emergency movement.
- CO2: Explain the effect of heat and toxic gases from fire on human.
- CO3: Interpret the mechanism of production of smoke due to fire, its harmful effect and movement within a building and the control measures to be adopted in buildings.
- CO4: Summarize the fire safety requirements for buildings of different occupancy as per the National Building Code of India
- CO5: Plan, design and distribute various fire safety measures required in buildings of different occupancy type.
- CO6: Explain the method of carrying out fire investigation, arson identification, fire training, fire safety audit and fire risk assessment.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Outline the importance of stream of people and stream width as parameter in design of evacuation route	3	3	L1	1	3
(b) Define relative density.	3	3	L1	1	6
(c) For a building with height up to 10 m define the open space requirement.	3	3	L1	5	3
(d) Explain the concept of fire zones.	3	3	L1	5	3
(e) List the requirements for detection zones for non-addressable automatic system with manual call point.	3	3	L1	4	3
(f) Outline the procedure for siting and spacing manual call point in buildings.	3	3	L2	5	3
(g) Outline the need for fire risk assessment for buildings.	3	3	L1	6	3
(h) Define Arson and list the motives for arson.	3	3	L1	6	3

PART B

(4 × 12 = 48)

II. (a) Outline the features of Emergency Evacuation and Panic Evacuation.	6	L1	1	6
(b) Compare specific traffic clearing capacity with traffic clearing capacity of an exit.	6	L3	1	6

OR

(P.T.O.)

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		Marks	BL	CO	PO
III.	(a) Determine the width, number of exits required, and aisle width for an auditorium having a capacity of 5000 people if it is required to be evacuated in 5 minutes.	6	L4	1	3
	(b) Summarise the method for calculation of building evacuation time based on travel distance.	6	L2	1	6
IV.	(a) Summarise the requirements for occupancies for life safety as per NBC.	6	L2	4	3
	(b) Outline the design criteria for internal stair and external stair as per NBC Part IV.	6	L2	4	3
OR					
V.	(a) Outline the design criteria for exit as per NBC part IV.	6	L2	5	3
	(b) Summarise the NBC guidelines for Fire lift, Fire Towers.	6	L2	5	3
VI.	(a) Outline the method to site and space heat detectors for pitched roof.	6	L2	5	3
	(b) List the requirements for non-addressable automatic detection system in buildings.	6	L2	5	3
OR					
VII.	(a) Summarise the requirements for addressable automatic systems.	6	L2	5	3
	(b) Outline the method to site and space Ember detectors as NBC.	6	L2	5	3
VIII.	(a) Summarise an action plan to control arson.	6	L2	6	3
	(b) Explain the concept of fire safety audit.	6	L2	6	3
OR					
IX.	Develop a sprinkler system design for various occupancy buildings.	12	L3	5	3

Bloom's Taxonomy Levels

L1 - 22.5%, L2 - 57.5%, L3 - 15%, L4 - 5%.
